PARTHA SARATHI SARKAR

CSE - AIML Student | Aspiring Data Scientist & ML Engineer

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PROFESSIONAL SUMMARY

Computer Science and Engineering (AI & ML) undergraduate building hands-on experience in Python, SQL, data analytics, and machine learning. Developed practical ML workflows spanning data preprocessing, exploratory analysis, supervised learning, model evaluation, and reproducible pipelines. Seeking Data Science / AI-ML opportunities to apply strong problem-solving skills to real-world data.

TECHNICAL SKILLS

Programming: Python, SQL, C, Java | Data: pandas, NumPy, Matplotlib, data cleaning, EDA
Machine Learning: scikit-learn, regression, classification, feature preprocessing, train/test splitting, model evaluation, Random Forest, pipelines
Tools: Git, GitHub, Joblib, TensorFlow (project exposure) | Core: Data Structures & Algorithms, problem solving

SELECTED PROJECTS

California Housing Price Prediction | *Python, scikit-learn, RandomForestRegressor, Joblib*

- Built a regression pipeline using an 80/20 stratified split based on income categories, reducing sampling bias while preserving the key distribution across train and test sets.
- Applied train-only preprocessing with SimpleImputer, StandardScaler, OneHotEncoder, and ColumnTransformer; evaluated on unseen test data with RMSE and persisted the complete pipeline with Joblib.

Iris Flower Classification | *Python, scikit-learn, supervised learning*

- Built a complete classification workflow on the Iris dataset, covering feature/target separation, model training, prediction, and evaluation.
- Used the project to strengthen practical understanding of classification, model validation, and the ML development lifecycle.

Plant Disease Detection | *Computer Vision, TensorFlow, ML*

- Worked on a leaf-image classification project and contributed to dataset preparation, model evaluation, and integration of the ML component into the larger application.
- Developed practical understanding of image-classification workflows and the constraints of integrating ML features into applications; implementation included AI-assisted development.

BharatFarm - Smart Agriculture Platform | *AI/ML, data-driven product concept*

- Contributed to a Smart India Hackathon project addressing farmer needs through plant-disease detection, weather alerts, price-risk prediction, smart mandi recommendations, and automated market linkage.
- Contributed to feature planning, technical solution design, presentation, and exploration of offline-use approaches for rural environments.

Data Analytics Projects | *Python, SQL, pandas, visualization*

- Completed a 5-project analytics portfolio spanning data cleaning, exploratory analysis, SQL querying, and visualization, including Netflix and pharmaceutical-sales analysis.
- Built a Python job-scraping project to collect structured job information and strengthen practical data-ingestion skills.

EDUCATION

B.Tech - Computer Science & Engineering (Artificial Intelligence & Machine Learning) | Undergraduate

PROJECT CONTRIBUTIONS

- Translated project ideas into concrete data/ML workflows, test cases, documentation, and presentations across academic and self-directed projects.
- Used AI-assisted/vibe-coding tools for implementation while taking responsibility for requirements, debugging, testing, data workflows, and understanding the underlying concepts; not presented as independent full-stack or JavaScript expertise.
- Practiced Data Structures & Algorithms alongside ML projects, strengthening analytical problem solving and coding fundamentals.

LANGUAGES

English | Bengali | Hindi | Japanese (developing proficiency)

CAREER FOCUS

Data Science | Machine Learning | Data Analytics | Computer Vision | AI Applications